Hafsa Usman

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Education

Qarshi University — Lahore, Pakistan
Bachelor of Science in Computer Science — *Oct. 2023 – Present*

Experience

Software Engineering Simulation — Aug. 2026 – Oct. 2026
Eynvision Labs — *Remote*

- Built an digital family memoir platform from scratch, handling everything from early design thinking and system architecture to the final MVP deployment using Next.js 15, FastAPI, and PostgreSQL.
- Presented design thinking reasoning during iterative review sessions and continuously refined the architecture based on feedback, ultimately resulting in our team's designs and codebase being selected as the top implementation and merged into production.
- Collaborated within an autonomous 3-person team by continuously discussing, discarding, and iterating on technical solutions, and conducted peer PR reviews to safely integrate code and keep development moving.
- Integrated the Gemini model to analyze and group unstructured memories into timeline chapters, and built an archive context builder allowing users to chat directly with their database records.
- Developed manual editing endpoints with an owner locking mechanism to protect user modifications from AI overwrites, and applied literal enums for accurate PostgreSQL date validation.

Projects

Spring Boot Backend Services | *Java, Spring Boot, Maven* — Jan. 2025 – Present

- Developed RESTful APIs using Java and Spring Boot following layered architecture principles.
- Implemented data persistence using Spring Data JPA while applying dependency injection and object-oriented design principles.

SSH Attack Detector | *Python, Scikit-learn* — 2026

- Built a machine learning application to identify anomalous SSH login behavior using K-Means clustering.
- Processed authentication logs, visualized clustering results, and evaluated model performance using Python and Scikit-learn.

Smart Slot: Desktop Management App | *Java, OOP* — Jan. 2024 – June 2024

- Developed a Java desktop application to automate academic timetable generation using object-oriented programming principles.
- Designed an additional finance management module to record and monitor personal expenses.

Mobile UI/UX Prototype (HCI Project) | *Figma* — Feb. 2026 – Present

- Designed a high-fidelity mobile application prototype to translate the core logic of the pharmacogenomics web platform into a clean mobile user interface.
- Applied Human-Computer Interaction (HCI) principles to construct user flows, interactive wireframes, and components.

Technical Skills

Languages: Python, TypeScript, Java, C++
Frameworks & Libraries: Next.js, FastAPI, React, Tailwind CSS, Spring Boot
Databases & Tools: PostgreSQL, Git, GitHub, Docker, Postman, Figma
Core Concepts: Full-Stack Development, LLM Integration, RESTful APIs, System Architecture

Leadership & Extracurriculars

Sub-Editor & Content Writer — March 2026 – Present
University Newsletter (Qarshi University) — *Lahore, Pakistan*

- Wrote articles covering university events and designed newsletter pages to maintain a consistent visual style.